

GARMIN®

AVIATION TRAINING

EFIS System Use Cases

AEA Nationals 2026

Garmin Aviation Training is intended to provide the best practices for the installation, use and maintenance of Garmin Aviation products. It is incumbent on technicians and pilots to ensure that the instructions contained in the most recent versions of Garmin product manuals, bulletins and notifications are followed.

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- ✈ In this course we'll break down the various intended use cases for a given EFIS system to satisfy your Certified Aircraft Customer.
- ✈ At the end of this the learner should understand the desired use case for the federated EFIS systems Garmin has.

✈ Paragraph B of the Subsection: The electronic display indicators, including their systems and installations, and considering other airplane systems, must be designed so that one display of information essential for continued safe flight and landing will remain available to the crew, without need for immediate action by any pilot for continued safe operation, after any single failure or probable combination of failures.



✈ Paragraph C of the Subsection: As used in this section, “instrument” includes devices that are physically contained in one unit, and devices that are composed of two or more physically separate units or components connected together (such as a remote indicating gyroscopic direction indicator that includes a magnetic sensing element, a gyroscopic unit, an amplifier, and an indicator connected together). As used in this section, “primary” display refers to the display of a parameter that is located in the instrument panel such that the pilot looks at it first when wanting to view that parameter.





✈ Paragraph A of Subsection: Installed systems must provide the flightcrew member who sets or monitors parameters for the flight, navigation, and powerplant, the information necessary to do so during each phase of flight. This information must:

- ✈ (1) Be presented in a manner that the crewmember can monitor the parameter and determine trends, as needed, to operate the airplane; and
- ✈ (2) Include limitations, unless the limitation cannot be exceeded in all intended operations.





✈ Paragraph B of Subsection: Indication systems that integrate the display of flight or powerplant parameters to operate the airplane or are required by the operating rules of this chapter must:

- ✈ (1) Not inhibit the primary display of flight or powerplant parameters needed by any flightcrew member in any normal mode of operation; and
- ✈ (2) In combination with other systems, be designed and installed so information essential for continued safe flight and landing will be available to the flightcrew in a timely manner after any single failure or probable combination of failures.

✈️ FAA Advisory Circular 25.11B offers the most comprehensive breakdown of Design, Installation, Integration, and Approval of Electronic Flight Deck displays, components, and systems installed in aircraft.

✈️ EFIS Requires:

✈️ Primary Flight Display

- ✈️ Attitude
- ✈️ Altitude
- ✈️ Airspeed
- ✈️ Magnetic Heading



✈ EFIS Requires:

✈ Primary Flight Display

- ✈ Attitude

- ✈ Altitude

- ✈ Airspeed

- ✈ Stabilized Heading

✈ Multifunction Display

- ✈ GPS Moving Map

- ✈ Pilot Situational Awareness Tools

 - ✈ Ancillary and Safety of Flight

✈ Engine Indication Systems



G5 Electronic Flight Instrument

- ✦ Attitude Indicator
- ✦ Turn coordinator
- ✦ Slip/Skid ball
- ✦ HSI page
- ✦ Entry point for Technically Advanced Aircraft Bare-minimum when paired with an IFR navigator and GFC 500 Digital Autopilot



- ✈️ This is made to provide essential instrumentation
- ✈️ Think replacing the Attitude Gyro as it fails
- ✈️ No frills
- ✈️ Limited integration
- ✈️ Ideal for VFR Aircraft



✈ The GI 275

✈ Displays

- ✈ Primary Attitude Indicator (Attitude, Altitude, Airspeed, and Heading) Standby Indicator
- ✈ Horizontal Situation Indicator (HSI) and Enhanced HSI
- ✈ Engine Indication System (EIS)
- ✈ Multi-Function Display (MFD)

✈ Full Authority EFIS requiring minimum panel modification

✈ Ideal for classic aircraft restoration



- ✈️ The GI275 is made to provide a more complete avionics integration experience with minimal panel modification
- ✈️ Think not only Upgrading analog instruments but also adding capability to the airframe
- ✈️ Easily Integrated with HSDB systems



✈️ The G3X

- ✈️ Primary Attitude Indicator (Attitude, Altitude, Airspeed, and Heading)
- ✈️ Horizontal Situation Indicator (HSI)
- ✈️ Engine Indication System (EIS)
- ✈️ Multi-Function Display (MFD)
- ✈️ UI designed by G1000 team meets the desire for a flightdeck experience in small aircraft
- ✈️ Features two sizes 10.6" and 7" Portrait
- ✈️ Ideal for aircraft which offer upgrade potential



✈️ The G3X

- ✈️ Made to provide a Flightdeck like experience
- ✈️ All integration can be done on one screen (REF: CNS Bar)
- ✈️ Think a customer who wants to improve their experience not just replace broken parts



✈️ The G5 is an Idel candidate as the Standby for a G3X

- ✈️ While the Airspeed and altitude are not primary indications they are suitable as a secondary indication
- ✈️ Because it communicates on the Can Bus it is designed to integrate with the G3X sytem



✈️ The TXi

- ✈️ Primary Attitude Indicator (Attitude, Altitude, Airspeed, and Heading)
- ✈️ Standby Attitude Indicator (Attitude, Altitude, Airspeed, and Heading)
- ✈️ Horizontal Situation Indicator (HSI) and HSI Map
- ✈️ Engine Indication System (EIS)
- ✈️ Multi-Function Display (MFD)
- ✈️ A range of sizes featuring the largest retrofit screen Garmin produces
 - ✈️ The GDU 1210, 1060, 700P and 700L are all different sizes and can be configured for layouts to provide a custom avionics package on approved aircraft.



✈ Provides a premium flightdeck experience

✈ All integration can be done on one screen

✈ Has the capability to integrate with a wide range of avionics both Garmin and non-Garmin (where necessary)

✈ Ideal for next generation aircraft integrations



- ✈ The GI 275 is a suitable Standby device for TXi
 - ✈ Primary qualified indications off all flight parameters
 - ✈ Communicates on the HSDB so is a fully featured standby for TXi





- ✈ Rugged Light VFR
- ✈ Minimal Panel
- ✈ Small aircraft Small mission
- ✈ Ideal candidate for a G5
- ✈ A small format navigator make this a full featured upgrade

- ✈ Classic airframe
- ✈ Space limited
- ✈ More integration opportunities
- ✈ Ideal candidate for GI 275
- ✈ Add a GTN 650Xi for Next Generation capabilities



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- ✈️ AN aircraft with Upgrade potential
 - ✈️ Engine STC's
- ✈️ Space availability for additional integration
- ✈️ Users tend to invest long-term
- ✈️ Ideal G3X Candidate
- ✈️ Add GTN and Small format for a full featured experience

- ✈ Larger Multi engine
- ✈ Candidate for a 135 Operation
- ✈ Possible existing equipment can be integrated
- ✈ Lots of panel space
- ✈ Three displays Two GTN's
Opportunity for a fully featured flight deck





What makes GI 275 a great option for classic airframes

- A. Rear mounted
- B. Fits a standard instrument cutout
- C. All in one HSI ADI



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What about an airframe would necessitate a TXi over a G3X

- A. Over 6000 MGTOW
- B. Customer wants GFC 500
- C. Multi Engine EIS Display
- D. Customer requested Synthetic Vision



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I attended EFIS System Use Cases in the Blue Room.

- A. True
- B. False



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- A. True
- B. False